Valid inequalities

Outline

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Learning objectives

At the end of this chapter you will be able to:

- define a valid inequality, and say when it cuts off a fractional point;
- derive a valid inequality from non-negative multipliers of the constraints by rounding down;
- improve a valid inequality until it touches a feasible integer point;
- measure the effect of a valid inequality on the optimal value of the LP relaxation and on the upper bound;
- explain why valid inequalities are the practical way to approach an ideal formulation.

Prerequisites

From this module:

- the chapter «Primal and dual bounds»: the LP relaxation, the rounded bound $\lfloor z(\mathrm{LP}) \rfloor$, and bounds from non-negative multipliers of the constraints
- the chapter «Ideal formulations»: formulations, the convex hull, and the comparison of formulations through their polyhedra

Where we were

- In the car example the ideal formulation gives the upper bound 14, equal to the lower bound, so it certifies that $(4, 1)$ is optimal
- But we found its four inequalities by looking at the convex hull, which is not possible in general

We now learn to **derive** such inequalities
from the constraints.

Why valid inequalities?

Introduction: Why valid inequalities?

- The ideal formulation gives the best possible upper bound, but in general we cannot compute it
- We see a practical alternative: add to the model a few well-chosen inequalities
- We close with a challenge on the car example

Tightening the LP relaxation

- We cannot compute the **ideal formulation** of a large ILP problem
- But we can add to the model a **few inequalities** that tighten the LP relaxation
- Each one removes some **fractional points** and keeps **all the integer ones**

The car example

$$
\begin{array}{rrcrcr}
\max & 3\,x_1 & + & 2\,x_2 & & \\
\text{subject to} & 10\,x_1 & + & 4\,x_2 & \leq & 45 \\
& 4\,x_1 & + & 14\,x_2 & \leq & 49 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

- The LP relaxation has the optimal solution $(\frac{7}{2}, \frac{5}{2})$, with $z(\mathrm{LP}) = \frac{31}{2}$: the upper bound is $\lfloor z(\mathrm{LP}) \rfloor = 15$
- The heuristic solution $(4, 1)$ gives the lower bound 14

A challenge

In the car example the upper bound is 15
and the lower bound is 14.

Is there an inequality,
satisfied by **all the integer points**,
that pushes the upper bound down to 14?

Summary

- The ideal formulation is out of reach for large ILP problems
- A few added inequalities can **tighten the LP relaxation**: they remove fractional points and keep all the integer ones
- In the car example we look for an inequality that lowers the upper bound from 15 to 14

Valid inequalities

Introduction: Valid inequalities

- We define a valid inequality, and what it means that it cuts off a point or that it is tight
- We see what happens to the feasible region, to the polyhedron and to the upper bound when a valid inequality is added
- These definitions are the language of the rest of the chapter, and of the cutting plane algorithms later in the course

Definition 1: of valid inequality

An inequality $\sum_{j=1}^{n} \alpha_j x_j \leq \beta$ *is a* ***valid inequality*** *for an ILP problem with feasible region* F *if it is satisfied by all the points of* F*.*

A valid inequality never removes a feasible solution.
What can it remove, then?
And how good can it be?

Definition 2: cut-off point and tight valid inequality

Let $\sum_{j=1}^{n} \alpha_j x_j \leq \beta$ be a valid inequality for an ILP problem with feasible region F.

- *The inequality **cuts off** a point $\boldsymbol{x}^* \in \mathbb{R}^n$ if*

$$\sum_{j=1}^{n} \alpha_j x_j^* > \beta.$$

- *The inequality is **tight** if at least one point of F satisfies it with equality.*

What a valid inequality cuts off

- A valid inequality cuts off **no point of** F
- But it can cut off **fractional points** of the LP relaxation
- The right-hand side of a **tight** inequality cannot be decreased without losing a feasible solution

Adding a valid inequality

- Adding a valid inequality to the formulation does **not change** F
- It **shrinks the polyhedron** P
- The new formulation is **stronger**, so its LP relaxation gives an upper bound that is not worse

Valid inequalities and the convex hull

- A valid inequality is satisfied by the points of F if and only if it is satisfied by their convex hull
- Hence it is valid for $\mathrm{conv}(F)$
- The inequalities of the **ideal formulation** are valid inequalities

The role of valid inequalities

A valid inequality removes **fractional points**
and never an integer feasible solution.

Adding valid inequalities makes the formulation
stronger and the upper bound **better**.

Example: valid inequalities of the car example

- The inequalities $x_1 \leq 4$ and $x_2 \leq 3$ of the previous chapters are valid inequalities of the car example
- They do not cut off the optimal solution $(\frac{7}{2}, \frac{5}{2})$ of the LP relaxation: $\frac{7}{2} \leq 4$ and $\frac{5}{2} \leq 3$
- So they do not improve the upper bound 15

Summary

- A **valid inequality** is satisfied by all the points of F
- It **cuts off** a point when the point violates it; it is **tight** when a point of F satisfies it with equality
- Adding it leaves F unchanged, shrinks P, and gives an upper bound that is not worse
- The inequalities of the ideal formulation are valid; $x_1 \leq 4$ and $x_2 \leq 3$ are valid but do not improve the bound 15

Deriving valid inequalities by rounding

Introduction: Deriving valid inequalities by rounding

- We need a way to build valid inequalities from the constraints, without looking at the convex hull
- We combine the constraints with non-negative multipliers, as for the dual bounds, and then round
- We try it on the car example, and see that cutting off the LP optimum is not always enough

Combine and round down

- The constraints are combined with **non-negative multipliers**, as for the dual bounds
- Since the variables are integer, the right-hand side of the combination can be **rounded down**

Proposition 1: valid inequalities by rounding

Let $\pi_1, \pi_2, \ldots, \pi_m \geq 0$ *be multipliers such that*

$$\sum_{i=1}^{m} \pi_i \, a_{ij} \geq \alpha_j, \quad \text{with } \alpha_j \in \mathbb{Z}, \qquad \forall j \in \{1, 2, \ldots, n\}.$$

Then the inequality

$$\sum_{j=1}^{n} \alpha_j \, x_j \;\leq\; \left\lfloor \sum_{i=1}^{m} \pi_i \, b_i \right\rfloor \tag{1}$$

is valid for the ILP problem.

Proof (1/2)

For every feasible solution:

$$\sum_j \alpha_j x_j \leq \sum_j \Big(\sum_i \pi_i a_{ij} \Big) x_j$$

$$= \sum_i \pi_i \Big(\sum_j a_{ij} x_j \Big)$$

$$\leq \sum_i \pi_i b_i$$

- the first inequality holds by $x_j \geq 0$, since $\sum_i \pi_i a_{ij} \geq \alpha_j$
- the second one holds by the constraints multiplied by $\pi_i \geq 0$

Proof (2/2)

- We have shown that, for every feasible solution,

$$\sum_j \alpha_j x_j \leq \sum_i \pi_i b_i$$

- The left-hand side is an **integer**, because α_j and x_j are integer
- So it is at most the integer part of the right-hand side:

$$\sum_j \alpha_j x_j \leq \left\lfloor \sum_i \pi_i b_i \right\rfloor$$

□

Integer and non-negative variables

- The proposition needs all the variables to be **integer** and **non-negative**
- With a continuous variable the left-hand side is not an integer, and the right-hand side cannot be rounded

We now apply the proposition to the car example, with the simplest multiplier: $\frac{1}{2}$.

Example: rounding with multiplier $\frac{1}{2}$ (1/3)

- In the car example, multiply the first constraint by $\frac{1}{2}$:

$$5\,x_1 + 2\,x_2 \leq \frac{45}{2}$$

- Since the left-hand side is integer, we obtain the valid inequality

$$5\,x_1 + 2\,x_2 \leq 22$$

- Multiplying the second constraint by $\frac{1}{2}$ we get $2\,x_1 + 7\,x_2 \leq \frac{49}{2}$, hence

$$2\,x_1 + 7\,x_2 \leq 24$$

Example: rounding with multiplier $\frac{1}{2}$ (2/3)

- Both inequalities cut off the optimal solution $(\frac{7}{2}, \frac{5}{2})$ of the LP relaxation:

$$5 \cdot \frac{7}{2} + 2 \cdot \frac{5}{2} = \frac{45}{2} > 22$$

$$2 \cdot \frac{7}{2} + 7 \cdot \frac{5}{2} = \frac{49}{2} > 24$$

- We add each of them to the formulation and solve the new LP relaxation

Example: rounding with multiplier $\frac{1}{2}$ (3/3)

The new LP relaxations give:

Added inequality	$\tilde{\boldsymbol{x}}(\mathrm{LP})$	$z(\mathrm{LP})$	$\lfloor z(\mathrm{LP}) \rfloor$
none	$(\frac{7}{2}, \frac{5}{2})$	$\frac{31}{2}$	15
$5\,x_1 + 2\,x_2 \leq 22$	$(\frac{105}{31}, \frac{157}{62})$	$\frac{472}{31}$	15
$2\,x_1 + 7\,x_2 \leq 24$	$(\frac{219}{62}, \frac{75}{31})$	$\frac{957}{62}$	15

- The LP optimum moves, but the upper bound does not improve: $\lfloor \frac{472}{31} \rfloor = \lfloor \frac{957}{62} \rfloor = 15$

Cutting off the LP optimum is not enough

A valid inequality that cuts off the optimal LP solution does **not always** improve the upper bound.

The choice of the multipliers matters:
we look for inequalities that **touch the integer points**.

Summary

- Combining the constraints with multipliers $\pi_i \geq 0$, with combined coefficients at least $\alpha_j \in \mathbb{Z}$, and **rounding down** the right-hand side gives a valid inequality
- It needs all the variables to be integer and non-negative
- In the car example, $5\,x_1 + 2\,x_2 \leq 22$ and $2\,x_1 + 7\,x_2 \leq 24$ cut off $(\frac{7}{2}, \frac{5}{2})$, but the upper bound stays at 15

Improving a valid inequality

Introduction: Improving a valid inequality

- Rounding gives a valid inequality, but its right-hand side can still be too large
- We lower the right-hand side step by step, until the inequality touches a feasible integer point
- On the car example this procedure closes the gap: it answers the challenge

The strongest right-hand side

- Given the coefficients α_j, the strongest valid inequality has the **smallest possible right-hand side**
- This right-hand side is the maximum of $\sum_j \alpha_j x_j$ over F
- We find it by lowering the right-hand side β of (1) until a feasible integer point reaches it

Lowering the right-hand side

Input: An ILP problem in canonical maximization form with $F \neq \emptyset$, and integer coefficients $\alpha_1, \alpha_2, \ldots, \alpha_n$.

Output: A tight valid inequality $\sum_{j=1}^{n} \alpha_j x_j \leq \beta$.

Step 0: **Coarse bound**

Find multipliers $\pi_i \geq 0$ with $\sum_i \pi_i a_{ij} \geq \alpha_j$ for all j, and set $\beta \leftarrow \lfloor \sum_i \pi_i b_i \rfloor$.

Step 1: **Test the right-hand side**

If no feasible integer point satisfies $\sum_j \alpha_j x_j = \beta$, set $\beta \leftarrow \beta - 1$ and repeat Step 1.

Step 2: **Tight inequality**

A feasible integer point satisfies $\sum_j \alpha_j x_j = \beta$, so $\sum_j \alpha_j x_j \leq \beta$ is valid and tight. `STOP`.

How to carry out the steps

- In Step 0 the multipliers can be chosen freely: the better they are, the fewer repetitions of Step 1 are needed
- In Step 1, for two variables:
  - we substitute one variable using the equation
  - we bound the other one with the constraints
  - if the bounds contradict each other, no point exists

We now apply the procedure to the car example,
with two directions α.

Example: the inequality $x_1 + x_2 \leq 5$ (1/4)

- We look for a valid inequality with $\alpha = (1, 1)$
- **Step 0.** With the multipliers $(\frac{1}{14}, \frac{1}{14})$ the sum of the constraints, divided by 14, is

$$x_1 + \frac{9}{7} x_2 \leq \frac{47}{7}$$

- The coefficients are at least $(1, 1)$, so

$$x_1 + x_2 \leq \left\lfloor \frac{47}{7} \right\rfloor = 6$$

and we set $\beta = 6$

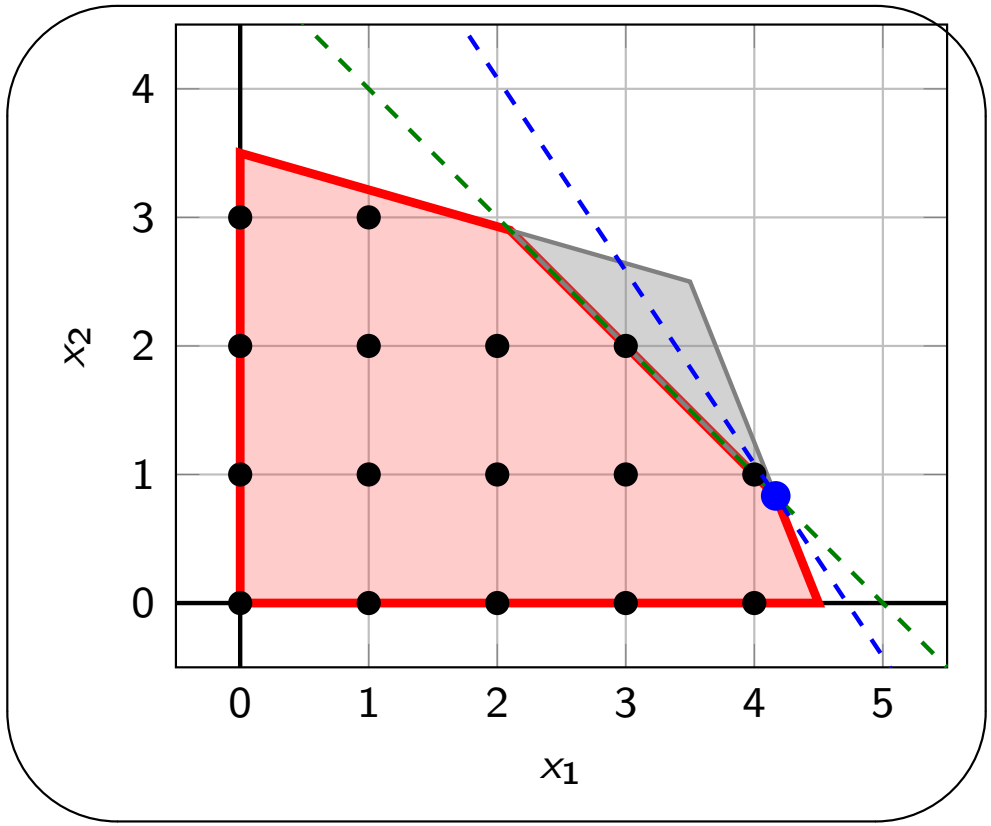
Example: the inequality $x_1 + x_2 \leq 5$ (2/4)

- **Step 1,** $\beta = 6$**.** Let $x_1 + x_2 = 6$, so $x_2 = 6 - x_1$
- The first constraint becomes $6\,x_1 + 24 \leq 45$, that is $x_1 \leq \frac{7}{2}$, hence $x_1 \leq 3$
- The second becomes $84 - 10\,x_1 \leq 49$, that is $x_1 \geq \frac{7}{2}$, hence $x_1 \geq 4$
- The bounds contradict each other: no feasible point has $x_1 + x_2 = 6$, and we set $\beta = 5$

Example: the inequality $x_1 + x_2 \leq 5$ (3/4)

- **Step 1,** $\beta = 5$. Let $x_2 = 5 - x_1$. The constraints become $x_1 \leq \frac{25}{6}$, hence $x_1 \leq 4$, and $x_1 \geq \frac{21}{10}$, hence $x_1 \geq 3$
- There is no contradiction, and the points $(3, 2)$ and $(4, 1)$ are feasible with $x_1 + x_2 = 5$
- **Step 2.** The inequality $x_1 + x_2 \leq 5$ is valid and tight
- It cuts off $(\frac{7}{2}, \frac{5}{2})$ (the sum is 6)
- The new LP relaxation has the optimal solution $(\frac{25}{6}, \frac{5}{6})$ with $z(\mathrm{LP}) = \frac{85}{6}$, so the upper bound is $\lfloor \frac{85}{6} \rfloor = 14$

Example: the inequality $x_1 + x_2 \leq 5$ (4/4)



The valid inequality $x_1 + x_2 \leq 5$ (green) cuts off the gray triangle, which contains no integer point, and the optimal solution of the previous LP relaxation. The new optimal solution is $(\frac{25}{6}, \frac{5}{6})$ (blue), with $z = \frac{85}{6}$ (dashed line).

Example: the inequality $x_1 + 2\,x_2 \leq 7$ (1/4)

- We look for a valid inequality with $\alpha = (1, 2)$
- **Step 0.** With the multipliers $(\frac{3}{62}, \frac{4}{31})$ the combination of the constraints is $x_1 + 2\,x_2 \leq \frac{17}{2}$, because

$$\frac{3 \cdot 10}{62} + \frac{4 \cdot 4}{31} = 1$$

$$\frac{3 \cdot 4}{62} + \frac{4 \cdot 14}{31} = 2$$

$$\frac{3 \cdot 45}{62} + \frac{4 \cdot 49}{31} = \frac{17}{2}$$

- Hence $x_1 + 2\,x_2 \leq \lfloor \frac{17}{2} \rfloor = 8$, and we set $\beta = 8$

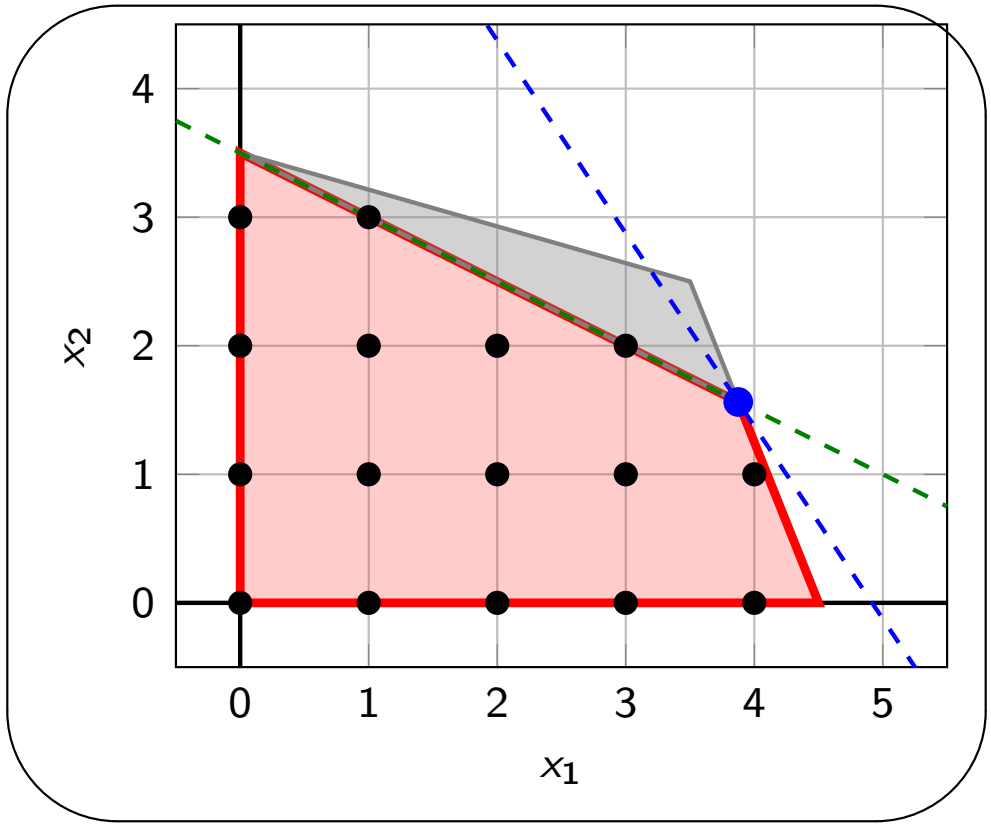
Example: the inequality $x_1 + 2\,x_2 \leq 7$ (2/4)

- **Step 1,** $\beta = 8$. Let $x_1 = 8 - 2\,x_2$
- The first constraint becomes $80 - 16\,x_2 \leq 45$, hence $x_2 \geq 3$
- The second becomes $32 + 6\,x_2 \leq 49$, hence $x_2 \leq 2$
- They contradict each other, and we set $\beta = 7$

Example: the inequality $x_1 + 2\,x_2 \leq 7$ (3/4)

- **Step 1,** $\beta = 7$**.** Let $x_1 = 7 - 2\,x_2$: the constraints become $70 - 16\,x_2 \leq 45$, hence $x_2 \geq 2$, and $28 + 6\,x_2 \leq 49$, hence $x_2 \leq 3$
- The points $(3, 2)$ and $(1, 3)$ are feasible with $x_1 + 2\,x_2 = 7$
- **Step 2.** The inequality $x_1 + 2\,x_2 \leq 7$ is valid and tight
- The new LP relaxation has the optimal solution $(\frac{31}{8}, \frac{25}{16})$ with $z(\mathrm{LP}) = \frac{59}{4}$, so the upper bound is $\lfloor \frac{59}{4} \rfloor = 14$

Example: the inequality $x_1 + 2\,x_2 \leq 7$ (4/4)



The valid inequality $x_1 + 2\,x_2 \leq 7$ (green) cuts off the gray triangle, which contains no integer point. The new optimal solution is $(\frac{31}{8}, \frac{25}{16})$ (blue), with $z = \frac{59}{4}$ (dashed line).

The car example is solved

With the valid inequality $x_1 + x_2 \leq 5$,
or with $x_1 + 2\,x_2 \leq 7$,
the upper bound drops from 15 to 14.

It equals the lower bound given by $(4, 1)$:
the solution $(4, 1)$ **is certified optimal**,
and the gap is zero.

Back to the ideal formulation

- The two inequalities, together with $x_1 \leq 4$ and $x_2 \leq 3$, are exactly the four inequalities of the ideal formulation of the previous chapter
- For the car example we have reached the **convex hull** with valid inequalities **derived from the constraints**

Summary

- Given α, the strongest right-hand side is the maximum of $\sum_j \alpha_j x_j$ over F
- We start from a rounded combination and lower β by one until a feasible integer point satisfies $\sum_j \alpha_j x_j = \beta$: the inequality is then **tight**
- In the car example, $x_1 + x_2 \leq 5$ and $x_1 + 2x_2 \leq 7$ give the upper bound 14: $(4, 1)$ is **certified optimal**
- Together with $x_1 \leq 4$ and $x_2 \leq 3$ they give the ideal formulation

Families of valid inequalities and cuts

Introduction: Families of valid inequalities and cuts

- With two variables we could choose the directions α by looking at the picture
- With many variables we need systematic ways to find valid inequalities
- We see the two sources used in practice, which are the subject of the next chapters

Two sources of valid inequalities

- With two variables the directions α are chosen on the picture. In general, valid inequalities come from two sources:

1. Families of inequalities for specific structures of the constraints, such as the **cover inequalities** of a constraint with positive coefficients of a binary problem (next chapter)

2. The rounding of this chapter, applied systematically to the rows of the optimal dictionary of the LP relaxation: the **Gomory fractional cuts** of the cutting plane algorithm, and then the **branch-and-cut** algorithm

Summary

- **Families** of valid inequalities exploit specific structures of the constraints: cover inequalities
- **Systematic rounding** of the rows of the optimal dictionary gives the Gomory fractional cuts
- They lead to the cutting plane algorithm and to branch-and-cut

Typical mistakes

Introduction: Typical mistakes

- We list the mistakes that are made most often with valid inequalities
- They concern the rounding, the multipliers and the effect on the bound
- Knowing them helps to derive correct and useful inequalities in the exercises

Typical mistakes

Observation 1: Typical mistakes (1/3)

- ***Rounding when a variable is continuous.*** *The rounding of (1) needs integer variables and integer coefficients α_j.*
- ***Rounding up instead of down.*** *The right-hand side is rounded down. Rounding up gives an inequality that is not valid, and it can remove integer solutions.*

Typical mistakes

Observation: Typical mistakes (2/3)

- ***Believing that every valid inequality improves the bound.*** *The inequalities* $5\,x_1 + 2\,x_2 \leq 22$ *and* $2\,x_1 + 7\,x_2 \leq 24$ *cut off the LP optimum, but leave the bound at* 15*.*
- ***Using multipliers that are not large enough.*** *The combined coefficients must be at least* α_j *for every* j*, and the multipliers must be non-negative.*

Typical mistakes

Observation: Typical mistakes (3/3)

- ***Stopping at a coarse right-hand side.*** *The inequality* $x_1 + x_2 \leq 6$ *is valid, but it does not cut off the LP optimum of the car example (its sum is* 6*); the tight inequality* $x_1 + x_2 \leq 5$ *does.*

Summary

Typical mistakes:

- Round only with integer variables and integer α_j, and always **down**
- Multipliers must be non-negative, with combined coefficients at least α_j
- Not every valid inequality improves the bound
- Lower a coarse right-hand side until the inequality is tight

In summary

Introduction: In summary

- We collect the key ideas of the chapter
- We close with the dashboard of the car example: the bounds found so far

Summary (1/3)

- A valid inequality is satisfied by all the feasible integer solutions; it can cut off fractional points of the LP relaxation, and it makes the formulation stronger
- Non-negative multipliers of the constraints, with the right-hand side rounded down, give valid inequalities for ILP problems
- Lowering the right-hand side until an integer point touches the line gives a tight inequality

Summary (2/3)

- A valid inequality helps only if it cuts off the LP optimum and improves the rounded bound
- In the car example, $x_1 + x_2 \leq 5$ or $x_1 + 2\,x_2 \leq 7$ lowers the upper bound from 15 to 14, and certifies that $(4, 1)$ is optimal

Summary (3/3)

Dashboard of the car example. The bounds on $z(\mathrm{ILP})$ found so far in this module:

Chapter	Lower bound	Upper bound	From
Primal and dual bounds	14	15	heuristic solution $(4,1)$, and $\lfloor z(\mathrm{LP}) \rfloor$ with $z(\mathrm{LP}) = \frac{31}{2}$
Ideal formulations	14	14	ideal formulation, optimal vertex $(4,1)$
Valid inequalities	14	14	$x_1 + x_2 \leq 5$ gives $z(\mathrm{LP}) = \frac{85}{6}$